

Sec.

a.

b)

Source

Source (0,0) can send through 2 lines $BW = 2.45613$
 $= 906B/s$

Host RAM $\xleftrightarrow{PCIe}$ TPU HBM
ICI = TPU to TPU network

Section 2 (Recorded)

1) v4p HBM BW = $1.2e12$

$200B \times 2 = 400$ billion bytes = $400e9$

$\therefore$ Total time = $\frac{400e9}{32} \div 1.2e12 = \frac{12.5e9}{1.2e12} \approx 10e-3$ seconds

$\approx 10ms$

2) v5e :

	v5e	v5p
CPU Hosts	32	2240
TPU Tensor Cores	256	17920
Total FLOPs/s	5216	4.1e18
Total HBM	4096 GB	860160 GB

$$v5e : a) \frac{16 \cdot 16}{4 \times 2} = 32$$

c) Assuming bf16, $16 \cdot 16 \cdot 1.97e14 = 5216$

d) $16^2 =$

$$v5p : a) \frac{16 \times 20 \times 28}{2 \times 2 \times 1} = 4 \cdot 20 \cdot 28 = 2240$$

$$b) 2 \cdot 16 \cdot 20 \cdot 28 = 1120 \times 16 = 17920$$

$$c) \text{Total FLOPs} = 16 \times 20 \times 28 \times 4.59e14 = 41126.4e14 = 4.11264e18$$

$$d) 96 \times 16 \times 20 \times 28 = 860160$$

3) Computation :

$$\text{FLOPs} = 2BD^2, \quad \text{FLOPs/s/v6e} = 9.20e14$$

$$\text{Throughput} = \frac{2BD^2}{9.20e14} = \frac{8BD^2}{9.20e14}$$

$$\text{Mem: } 2BD + 2DF + 2BF \quad \text{PCIe BW} = 1.6e10$$

$$\text{Throughput} = \frac{2BD + 2DF + 2BF}{1.6e10} = \frac{BD + DF + BF}{8e9}$$

$$= \frac{DF}{8e9} = \frac{4D}{8e9} = \frac{4D}{8e9}$$

$$T_{\text{math}} > T_{\text{comms}} \Rightarrow \frac{880^x}{9.2e14} > \frac{B}{2e9}$$

$$\Rightarrow B > \frac{9.2e14}{16e9} = 5.75e4 \approx 57500 //$$

4) a) FLOPs = (2)(B)(4096)(16384) = 134217728 B = $1.3 \times 10^8 B$

FLOPs for v5e = $3.94e14$

$$T_{\text{math}} = \frac{1.3 \times 10^8 B}{3.94e14}$$

$$T_{\text{mem}} = \frac{(16384 \cdot 4096 + 4096B + 16384B)}{8.2211}$$

$$\approx \frac{(6.7 \times 10^7 + 20000B)}{8.2211}$$

$$\max(T_{\text{math}}, T_{\text{mem}}) = \max\left(\frac{1.3 \times 10^8}{3.94e14} B, \frac{6.7 \times 10^7 + 20000B}{8.2211}\right)$$

↳ fnc of B

We would be compute-bound if $T_{\text{math}} > T_{\text{mem}}$

$$\Rightarrow \frac{1.3 \times 10^8}{5.94 \times 10^{14}} B > \frac{6.7 \times 10^7 + 20000B}{8.2 \times 10^{11}}$$

$$\Rightarrow 3.29 \times 10^{-7} B - 2.44e-8 B > 8.17e-5$$

$$\Rightarrow 3.046e-7 B > 8.17e-5$$

$$B > 268 \text{ to be compute bound!} //$$

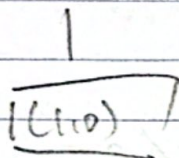
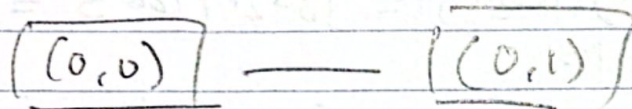
b) If loading is 22x faster, take $T_{\text{mem}}' = T_{\text{mem}} \times \frac{1}{22}$

$$\Rightarrow B > \frac{268}{22} \text{ to be compute bound now.}$$

5) a) rows = 3 cols = 3

$$3+3 = 6 \mu s$$

b) Total bytes = $2 \cdot 8 \cdot 128 \cdot 8192 = 1.7e7$ bytes

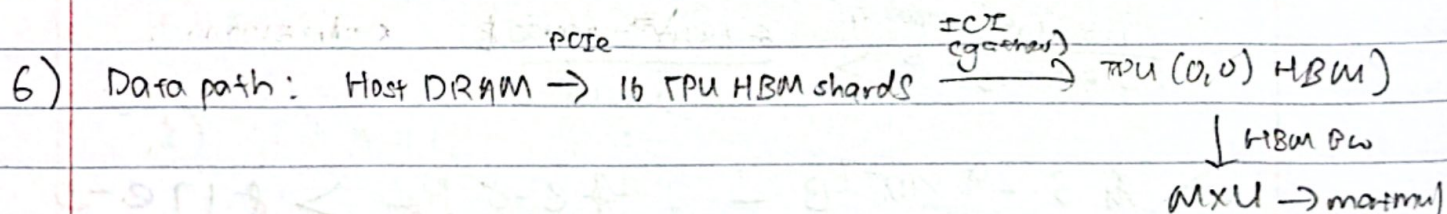


$\Rightarrow$ can send data 2-ways

$$\text{Total BW} = 2 \times 4.5e10 = 9.0e10$$

$$\therefore \text{Total time} = \frac{1.7e7}{9.0e10} = 188 \mu s$$

$$\text{Total time taken} = 188 + 6 = 194 \mu s$$



$$A = 17 \text{ GB}, N = 2 \text{ MB}$$

$\Rightarrow$ Each of the 16 GPUs hold $\sim 1 \text{ GB}$ (assume 1 GB)

i) As assumed in (Q3), PCIe BW = $1.6e10$ bytes/s = 16 GB/s

Since we have 16 PCIe links, total BW = $16 \times 16 = 256 \text{ GB/s}$

$$\therefore \text{Total time for PCIe} = \frac{16}{256} = \frac{1}{16} = 0.0625 \text{ s} = 62.5 \text{ ms}$$

ii) CPU must receive the remaining 15GB from other CPUs.

Each link 45GB/s $\Rightarrow$ Two links = 90GB/s

$$\text{Time} = \frac{15}{90} = \frac{1}{6} = 0.167\text{s} = 167\text{ms}$$

iii) HBM $\rightarrow$ MXU : Bytes = 16GB, HBM BW = 8.2GB/s

$$\text{Time} = \frac{16 \times 10^9}{8.2 \times 10^9} = 2 \times 10^{-2} = 20 \times 10^{-3} \text{s} = 20\text{ms}$$

iv) Compute : $2 \times 8 \times (128 \times 1024)^2 = 2.75 \times 10^{11}$ FLOPs

$$\text{Time} = \frac{2.75 \times 10^{11}}{1.97 \times 10^{14}} = 1.4 \times 10^{-3} = 1.4\text{ms}$$

i) 62.5ms ii) 167ms iii) 20ms iv) 1.4ms

Total $\approx \max(\text{i}, \text{ii}, \text{iii}, \text{iv}) = 167\text{ms}$ (I believe it's a lower bound, assuming perfect overlap and pipelining)